
GENERAL RELATIVITY

Tutorial problem set 9, 13.11.2020, with solutions

■ PROBLEM 25 Riemann normal coordinates.

By definition of a Riemann manifold, at a given point p there exists a coordinate system with respect to which the first partial derivatives of the metric components vanish (local flatness). In these coordinates (*Riemann normal coordinates* with respect to the point p), which by definition are the coordinates of a local inertial frame, the equation of a geodesic through the point p is $x^\mu = a^\mu s$, where $a^\mu = \left. \frac{dx^\mu}{ds} \right|_p$ and s is the curve parameter.

(a) Derive (from the geodesic equation) that

$$\Gamma_{\mu\nu}^\alpha \Big|_p = 0 , \quad (25.1)$$

and that

$$\partial_{(\beta} \Gamma_{\mu\nu)}^\alpha \Big|_p = 0 . \quad (25.2)$$

(b) Use identity (25.2) to show that the metric around point p can be written as

$$g_{\mu\nu} = g_{\mu\nu} \Big|_p + \frac{1}{3} R_{\alpha\mu\nu\beta} \Big|_p x^\alpha x^\beta + \mathcal{O}(x^3) . \quad (25.3)$$

where $x^\alpha \Big|_p = 0$.

Hints: First prove that (25.2) implies

$$\partial_\nu \Gamma_{\alpha\beta}^\mu = -\frac{1}{3} \left(R^\mu{}_{\beta\alpha\nu} + R^\mu{}_{\alpha\beta\nu} \right) , \quad (25.4)$$

and that this implies

$$\partial_\alpha \partial_\beta g_{\mu\nu} = -\frac{1}{3} \left(R_{\mu\alpha\nu\beta} + R_{\mu\beta\nu\alpha} \right) . \quad (25.5)$$

SOLUTION: (a) Applying the geodesic equation,

$$\frac{d^2 x^\mu}{ds^2} + \Gamma_{\alpha\beta}^\mu \frac{dx^\alpha}{ds} \frac{dx^\beta}{ds} = 0 \quad (25.6)$$

to $x^\mu(s) = a^\mu s$ at point p one obtains,

$$\left[\Gamma_{\alpha\beta}^\mu \right]_p a^\alpha a^\beta = 0 . \quad (25.7)$$

Since a^α are *a priori* arbitrary vectors at p , we conclude that $\left[\Gamma_{\alpha\beta}^\mu \right]_p = 0$.

(b) Taking one more parametric derivative of the geodesic equation (25.6), $\frac{d}{ds} = \frac{dx^\nu}{ds} \partial_\nu$ gives,

$$\left[\partial_\nu \Gamma_{\alpha\beta}^\mu \right]_p a^\nu a^\alpha a^\beta = 0 . \quad (25.8)$$

where we took account of (25.7). Eq. (25.2) then follows upon realising that the factors $a^\nu a^\alpha a^\beta$ symmetrize $[\partial_\nu \Gamma_{\alpha\beta}^\mu]_p$ in (25.8).

To show (25.4) note first that at point p (at which Christoffel symbols vanish),

$$R_{\nu\alpha\beta}^\mu + R_{\alpha\nu\beta}^\mu = \partial_\alpha \Gamma_{\beta\nu}^\mu - \partial_\beta \Gamma_{\alpha\nu}^\mu + \partial_\nu \Gamma_{\beta\alpha}^\mu - \partial_\beta \Gamma_{\nu\alpha}^\mu = -3\partial_\beta \Gamma_{\nu\alpha}^\mu, \quad (25.9)$$

where the last equality follows from $\partial_{(\nu} \Gamma_{\alpha\beta)}^\mu = 0$ at p . Eq. (25.4) follows trivially from (25.9)

Upon lowering the index μ in $\partial_\beta \Gamma_{\nu\alpha}^\mu$ and renaming some indices, Eq. (25.9) can be recast as (again at point p),

$$g_{\rho\mu} \partial_\nu \Gamma_{\alpha\beta}^\rho = \frac{1}{3} (R_{\mu\alpha\nu\beta} + R_{\mu\beta\nu\alpha}) = g_{\rho(\mu} \partial_{\nu)} \Gamma_{\alpha\beta}^\rho, \quad (25.10)$$

where the second equality follows from the symmetry of the Riemann tensor under the exchange of the first and second pair of indices.

In order to show (25.5), note that the metric compatibility $\nabla_\alpha g_{\mu\nu} = 0$ implies,

$$\begin{aligned} \nabla_\alpha \nabla_\beta g_{\mu\nu} &= \partial_\alpha \nabla_\beta g_{\mu\nu} = \partial_\alpha [\partial_\beta g_{\mu\nu} - \Gamma_{\beta\mu}^\rho g_{\rho\nu} - \Gamma_{\beta\nu}^\rho g_{\rho\mu}] \\ &= \partial_\alpha \partial_\beta g_{\mu\nu} - (\partial_\alpha \Gamma_{\beta\mu}^\rho) g_{\rho\nu} - (\partial_\alpha \Gamma_{\beta\nu}^\rho) g_{\rho\mu} = 0, \end{aligned} \quad (25.11)$$

where the first equality follows from the fact that at point p Christoffel symbols vanish and the third equality follows from the observation that all first derivatives of $g_{\mu\nu}$ vanish at p . Eq. (25.11) implies,

$$\begin{aligned} \partial_\alpha \partial_\beta g_{\mu\nu} &= (\partial_\alpha \Gamma_{\beta\mu}^\rho) g_{\nu\rho} + (\partial_\alpha \Gamma_{\beta\nu}^\rho) g_{\mu\rho} = (\partial_{(\alpha} \Gamma_{\beta)\mu}^\rho) g_{\nu\rho} + (\partial_{(\alpha} \Gamma_{\beta)\nu}^\rho) g_{\mu\rho} \\ &= -\frac{1}{2} [g_{\rho\nu} (\partial_\mu \Gamma_{\alpha\beta}^\rho) + g_{\rho\mu} (\partial_\nu \Gamma_{\alpha\beta}^\rho)] = -g_{\rho(\mu} \partial_{\nu)} \Gamma_{\alpha\beta}^\rho, \end{aligned} \quad (25.12)$$

where the second equality follows from, $\nabla_\alpha \nabla_\beta g_{\mu\nu} = \nabla_\beta \nabla_\alpha g_{\mu\nu}$ and the third equality follows from $\partial_{(\alpha} \Gamma_{\beta\gamma)}^\mu = 0$. Comparing Eq. (25.12) with (25.10) we see that,

$$\partial_\alpha \partial_\beta g_{\mu\nu} = -g_{\rho(\mu} \partial_{\nu)} \Gamma_{\alpha\beta}^\rho = -\frac{1}{3} (R_{\mu\alpha\nu\beta} + R_{\mu\beta\nu\alpha}) = \frac{1}{3} (R_{\alpha\mu\nu\beta} + R_{\beta\mu\nu\alpha}), \quad (25.13)$$

where we made use of the fact that $g_{\rho\mu} \partial_\nu \Gamma_{\alpha\beta}^\rho = g_{\rho(\mu} \partial_{\nu)} \Gamma_{\alpha\beta}^\rho$.

As a final step, note that the metric tensor can be Taylor expanded around p as,

$$\begin{aligned} g_{\mu\nu}(x) &= [g_{\mu\nu}]_p + [\partial_\alpha g_{\mu\nu}]_p x^\alpha + \frac{1}{2} [\partial_\alpha \partial_\beta g_{\mu\nu}]_p x^\alpha x^\beta + \mathcal{O}(x^3) \\ &= \eta_{\mu\nu} + \frac{1}{3} R_{\alpha\mu\nu\beta} x^\alpha x^\beta + \mathcal{O}(x^3), \end{aligned} \quad (25.14)$$

where we assumed that $x^\mu = 0$ at point p and to get the last equality we made use of, $R_{\alpha\mu\nu\beta} x^\alpha x^\beta = R_{\beta\mu\nu\alpha} x^\alpha x^\beta$.

■ PROBLEM 26 Energy-momentum tensor for a scalar field.

The action for the scalar field non-minimally coupled to curvature is

$$S_\phi = \int d^4x \sqrt{-g} \left(-\frac{1}{2} g^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - V(\phi) - \frac{1}{2} \xi R \phi^2 \right), \quad (26.1)$$

where ξ is the so-called non-minimal coupling. This is one of the models used in inflationary theories, which goes under the name *Higgs inflation*.

- (a) Derive the equation of motion for the scalar field.
- (b) Calculate the energy-momentum tensor for the minimally coupled scalar field ($\xi=0$).
- (c) Calculate the additional contribution to the energy-momentum tensor coming from the non-minimal coupling of the scalar.

Recall that the definition of the energy-momentum tensor is

$$T_{\mu\nu}(x) = \frac{-2}{\sqrt{-g(x)}} \frac{\delta S_\phi}{\delta g^{\mu\nu}(x)}. \quad (26.2)$$

SOLUTION: (a) Varying the action (26.1) with respect to $\phi(x')$ gives,

$$\frac{\delta S_\phi}{\delta \phi(x')} = \int d^4x \sqrt{-g} \left[-g^{\mu\nu} (\partial_\mu \phi) \partial_\nu \delta^4(x - x') - \frac{dV(\phi)}{d\phi} \delta^4(x - x') - \xi R \phi \delta^4(x - x') \right], \quad (26.3)$$

where we made use of the fundamental relation of variational calculus, $\delta \phi(x)/\delta \phi(x') = \delta^4(x - x')$. The first term in (26.3) ought to be integrated by parts to obtain,

$$\begin{aligned} \frac{\delta S_\phi}{\delta \phi(x')} &= \int d^4x \sqrt{-g} \left\{ \frac{1}{\sqrt{-g}} \partial_\nu [\sqrt{-g} g^{\mu\nu} (\partial_\mu \phi)] - \frac{dV(\phi)}{d\phi} - \xi R \phi \right\} \delta^4(x - x') \\ &\quad - \int d^4x \partial_\nu \{ \sqrt{-g} g^{\mu\nu} (\partial_\mu \phi) \delta^4(x - x') \}. \end{aligned} \quad (26.4)$$

The last term is a boundary term, which vanishes as long as the initial and final times of the integral differ from t' . The remaining terms can be integrated over x to give,

$$\begin{aligned} \frac{1}{\sqrt{-g}} \frac{\delta S_\phi}{\delta \phi(x')} &= \square' \phi(x') - \frac{dV(\phi(x'))}{d\phi} - \xi R \phi(x') = 0 \\ \square' \phi(x') &\equiv \frac{1}{\sqrt{-g(x')}} \partial'_\nu [\sqrt{-g} g^{\mu\nu}(x') (\partial'_\mu \phi(x'))] \end{aligned} \quad (26.5)$$

This is the Euler-Lagrange equation for ϕ . Note that equation (26.5) is a scalar with respect to general coordinate transformations.

(b) From Eq. (26.2) we see that we ought to vary (26.1) with respect to $g^{\mu\nu}(x')$. When $\xi = 0$ the variation gives,

$$\begin{aligned} \left[\frac{\delta S_\phi}{\delta g^{\mu\nu}(x')} \right]_{\xi=0} &= \int d^4x \left\{ -\frac{1}{2} \sqrt{-g} \delta_{(\mu}^\alpha \delta_{\nu)}^\beta \delta^4(x - x') (\partial_\alpha \phi) (\partial_\beta \phi) \right. \\ &\quad \left. - \frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta^4(x - x') \left[-\frac{1}{2} g^{\alpha\beta} (\partial_\alpha \phi) \partial_\beta \phi - V(\phi) \right] \right\}, \end{aligned} \quad (26.6)$$

where we made use of, $\delta \sqrt{-g} = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}$. Eq. (26.6) can be integrated over x to obtain,

$$[T_{\mu\nu}(x)]_{\xi=0} = (\partial_\mu \phi(x)) (\partial_\nu \phi(x)) + g_{\mu\nu}(x) \left[-\frac{1}{2} g^{\alpha\beta}(x) (\partial_\alpha \phi(x)) \partial_\beta \phi(x) - V(\phi(x)) \right], \quad (26.7)$$

where for simplicity after the integration we set $x' \rightarrow x$ and we made use of (26.2). Notice that the term in the square brackets is the Lagrangian density $\mathcal{L}_\phi$.

(c) To vary the non-minimally coupled term (26.6) we need the first variation of the Ricci tensor. From the lecture we know that,

$$\delta R_{\mu\nu} = \nabla_\rho (\delta \Gamma_{\nu\mu}^\rho) - \nabla_\nu (\delta \Gamma_{\rho\mu}^\rho) . \quad (26.8)$$

On the other hand $\delta \Gamma_{\nu\mu}^\rho$ can be written covariantly as (show that!),

$$\delta \Gamma_{\nu\mu}^\rho = - \left[\delta_\alpha^\rho g_{\beta(\mu} \nabla_{\nu)} - \frac{1}{2} g_{\alpha(\mu} g_{\nu)\beta} g^{\rho\sigma} \nabla_\sigma \right] \delta g^{\alpha\beta} . \quad (26.9)$$

Inserting this into (26.8) gives,

$$\delta R_{\mu\nu} = \left[-g_{\beta(\mu} \nabla_{\nu)} \nabla_\alpha + \frac{1}{2} g_{\alpha(\mu} g_{\nu)\beta} \square + \frac{1}{2} g_{\alpha\beta} \nabla_\mu \nabla_\nu \right] \delta g^{\alpha\beta} . \quad (26.10)$$

From here we easily see that variation of the Ricci scalar gives,

$$\delta R = (\delta g^{\mu\nu}) R_{\mu\nu} + g^{\mu\nu} [-\nabla_\beta \nabla_\alpha + g_{\alpha\beta} \square] \delta g^{\alpha\beta} . \quad (26.11)$$

Upon combining the contributions from δR with that from $\delta\sqrt{-g}$ one obtains the following contribution to the energy-momentum tensor from the non-minimally coupled term,

$$[T_{\mu\nu}(x)]_\xi = \xi \left[R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R + (g_{\mu\nu} \square - \nabla_\mu \nabla_\nu) \right] \phi^2 . \quad (26.12)$$

The total energy-momentum tensor is then the sum of the two contributions: (26.7) and (26.12)